HEALTHCARE CAPACITY & CARE LOAD ANALYTICS USING INTERACTIVE DASHBOARD AND LINEAR REGRESSION FORECASTING

Research Paper

Submitted By

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Submitted To	**Project Duration**
UMPL Internship Pro	**July 2026 – August 2026**

Project Duration

July 2026 – August 2026

Tools & Technologies Used

- **Python** **Pandas**
- **NumPy** **Matplotlib**
- **Seaborn** **Streamlit**
- **Scikit-learn (Linear Regression)** **Jupyter Notebook**

Dataset Used

Children in HHS Care Dataset

Submission Date

August 2026

Abstract

Healthcare organizations require continuous monitoring of patient care capacity to ensure efficient resource allocation and effective service delivery. This project presents an interactive Healthcare Capacity and Care Load Analytics system developed using Python, Streamlit, Pandas, Plotly, and statistical forecasting techniques.

The study analyzes healthcare data related to children in CBP custody and HHS care by performing exploratory data analysis (EDA), Key Performance Indicator (KPI) evaluation, trend analysis, and future demand forecasting. Interactive dashboards allow users to monitor healthcare capacity, compare CBP and HHS care loads, evaluate transfers and discharges, and explore historical trends through dynamic visualizations.

To enhance decision-making, an ARIMA-based forecasting model was implemented to predict future healthcare demand. The developed Streamlit web application integrates all analytical components into a user-friendly interface, enabling stakeholders to perform real-time exploration of healthcare data.

The project demonstrates how data analytics and forecasting can improve operational planning, resource utilization, and strategic decision-making in healthcare systems. The proposed framework provides an effective analytical solution that supports evidence-based policy planning and future healthcare capacity management.

Keywords:

Healthcare Analytics, Streamlit, Python, Data Visualization, Forecasting, ARIMA, KPI Analysis, Dashboard, Healthcare Capacity.

<u>Table of Content</u>

1. Introduction

Healthcare systems generate large volumes of operational data every day. Efficient utilization of this information enables organizations to monitor healthcare capacity, improve patient care, optimize resource allocation, and support strategic planning.

The care of unaccompanied children requires continuous coordination between multiple healthcare agencies, including Customs and Border Protection (CBP) and the Department of Health and Human Services (HHS). Monitoring patient intake, transfers, discharges, and healthcare capacity is essential for ensuring timely services and efficient utilization of available resources.

Traditional reporting methods often rely on static spreadsheets that provide limited analytical capabilities. These approaches make it difficult to identify long-term trends, monitor key performance indicators, and forecast future healthcare demand.

To address these challenges, this project develops an interactive Healthcare Capacity and Care Load Analytics Dashboard using Python and Streamlit. The system combines exploratory data analysis (EDA), KPI monitoring, statistical forecasting, and interactive visualizations into a unified analytical platform.

The dashboard enables healthcare administrators and decision-makers to analyze historical trends, compare healthcare metrics, evaluate operational performance, and forecast future healthcare demand using ARIMA forecasting techniques.

The developed solution demonstrates the practical application of modern data analytics technologies in healthcare decision support and capacity planning.

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2. Problem Statement

Healthcare organizations managing the care of unaccompanied children face significant challenges in monitoring healthcare capacity, patient movement, and resource utilization. Large volumes of daily operational data make it difficult to identify trends, evaluate system performance, and make timely decisions using traditional reporting methods.

The absence of interactive analytical tools limits the ability of decision-makers to monitor changes in healthcare demand, compare care load across agencies, and forecast future capacity requirements. As a result, resource planning, operational efficiency, and service delivery may be affected.

This project addresses these challenges by developing an interactive Healthcare Capacity and Care Load Analytics Dashboard that provides real-time visualization, KPI monitoring, comparative analysis, and demand forecasting. The system enables healthcare administrators to make data-driven decisions for effective capacity planning and resource management.

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3. Project Objectives

3.1 Primary Objectives

- Develop an interactive healthcare analytics dashboard for monitoring healthcare capacity.
- Analyze historical trends in children under CBP custody and HHS care.
- Evaluate key healthcare performance indicators (KPIs).
- Compare CBP custody, HHS care, transfers, and discharge activities.
- Forecast future healthcare demand using the ARIMA time series forecasting model.

3.2 Secondary Objectives

- Perform exploratory data analysis (EDA) to identify trends and patterns.
- Provide interactive visualizations for effective decision-making.
- Enable dynamic filtering and real-time dashboard exploration.
- Support healthcare administrators in resource planning.
- Improve operational transparency through data-driven insights.

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4. Dataset Description

The dataset used in this project contains daily records related to healthcare capacity and care management for unaccompanied children. The data captures operational activities such as children under CBP custody, children in HHS care, transfers, discharges, and overall system capacity. These records are used for exploratory analysis, KPI evaluation, trend monitoring, and forecasting.

Dataset Attributes

Attribute	Description
Date	Daily observation date
Children Apprehended and Placed in CBP Custody	Number of children entering CBP custody
Children in CBP Custody	Daily count of children in CBP custody
Children Transferred to HHS Custody	Number of children transferred to HHS
Children in HHS Care	Daily count of children receiving HHS care
Children Discharged from HHS Care	Number of children discharged from HHS care

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Dataset Characteristics

- **Dataset Type:** Time Series Healthcare Dataset
- **Data Format:** CSV
- **Time Granularity:** Daily Records
- **Analysis Period:** Historical Healthcare Records
- **Domain:** Healthcare Analytics
- **Purpose:** Capacity Monitoring, Trend Analysis, KPI Evaluation, and Forecasting

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5. Research Methodology

The research methodology followed in this project focuses on transforming raw healthcare data into meaningful insights through exploratory data analysis, KPI evaluation, interactive dashboard development, and forecasting techniques. The complete workflow was implemented using Python, Streamlit, Pandas, Plotly, and statistical forecasting methods.

The methodology consists of the following stages:

5.1 Data Collection

The healthcare dataset was collected in CSV format containing daily operational records related to unaccompanied children under the custody of Customs and Border Protection (CBP) and the Department of Health and Human Services (HHS).

The dataset includes information such as:

- Date
- Children Apprehended and Placed in CBP Custody
- Children in CBP Custody
- Children Transferred to HHS Custody
- Children in HHS Care
- Children Discharged from HHS Care

These variables provide a comprehensive overview of healthcare capacity and patient movement.

5.2 Data Preprocessing

Before performing analysis, the dataset was cleaned and prepared to improve data quality.

The preprocessing steps included:

- Handling missing values
- Removing duplicate records
- Converting date columns into datetime format
- Correcting inconsistent data types
- Creating additional calculated metrics for analysis
- Preparing data for visualization and forecasting

These preprocessing steps ensured that the dataset was accurate, consistent, and suitable for analysis.

5.3 Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand the characteristics and patterns within the dataset.

The analysis included:

- Summary statistics
- Distribution analysis
- Monthly trend analysis
- Correlation analysis
- Outlier identification
- Time-series visualization

EDA helped identify important healthcare trends and operational patterns before developing the dashboard.

5.4 Key Performance Indicator (KPI) Analysis

Several KPIs were calculated to evaluate healthcare system performance.

The major KPIs include:

- Total Children Apprehended
- Total Children Discharged
- Average HHS Care
- Maximum HHS Care
- Minimum HHS Care
- Net Intake
- Monthly Healthcare Capacity
- Transfer Performance

These indicators provide decision-makers with a quick overview of healthcare system performance.

5.5 Interactive Dashboard Development

An interactive healthcare analytics dashboard was developed using Python and Streamlit.

The dashboard includes:

- KPI Cards
- Trend Analysis
- Monthly Healthcare Monitoring
- CBP vs HHS Comparison
- Transfer vs Discharge Analysis
- Interactive Filters
- Dataset Preview
- Download Option
- Dynamic Charts using Plotly

The dashboard allows users to interactively explore healthcare data and monitor system performance.

5.6 Forecasting Analysis

Time-series forecasting was performed using the ARIMA (AutoRegressive Integrated Moving Average) model.

The forecasting process included:

- Historical trend analysis
- Moving average calculation
- ARIMA model implementation
- Future healthcare demand prediction
- Forecast visualization

The forecasting model assists healthcare administrators in estimating future healthcare capacity requirements.

5.7 Streamlit Web Application

The complete analytical system was deployed as an interactive Streamlit web application.

The application consists of four major modules:

- Dashboard
- KPI Analysis
- Forecasting
- About Project

The Streamlit application enables users to explore healthcare data through an intuitive web interface.

5.8 Tools and Technologies

The following tools and technologies were used throughout the project:

Tool	Purpose
Python	Data Analysis & Development
Streamlit	Interactive Web Application
Pandas	Data Manipulation
Plotly	Interactive Visualization
NumPy	Numerical Computation
Statsmodels (ARIMA)	Time Series Forecasting
VS Code	Development Environment
CSV Dataset	Data Source

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6. Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was performed to understand the structure, quality, and behavior of the healthcare dataset before developing the interactive dashboard and forecasting model. The objective of EDA was to identify trends, relationships, and performance patterns within the healthcare system.

The analysis was conducted using Python libraries such as Pandas, Plotly, and NumPy.

6.1 Summary Statistics

The first step involved generating descriptive statistics to understand the overall characteristics of the dataset.

The analysis included:

- Mean
- Median
- Standard Deviation
- Minimum Value
- Maximum Value
- Count of Records

These statistics provided a general overview of healthcare capacity and operational performance.

2) Statistical Summary

Descriptive statistics provide an overview of the distribution of numerical variables.

```
[16]: df.describe().T
```

[16]:

	count	mean	min	25%	50%	75%	max	std
Date	720	2024-07-06 05:29:59.999999744	2023-01-12 00:00:00	2023-10-16 18:00:00	2024-07-05 12:00:00	2025-03-25 06:00:00	2025-12-21 00:00:00	NaN
Children apprehended and placed in CBP custody*	720.0	93.523611	0.0	12.0	99.0	147.25	333.0	72.646625
Children in CBP custody	720.0	171.494444	7.0	36.0	193.0	263.25	531.0	126.354965
Children transferred out of CBP custody	720.0	128.668056	0.0	14.0	157.0	199.25	440.0	97.322012
Children in HHS Care	720.0	6061.275	1972.0	2467.75	6406.5	8010.25	11516.0	2833.070109
Children discharged from HHS Care	720.0	173.406944	0.0	19.75	181.0	267.0	505.0	125.702841
Year	720.0	2024.0125	2023.0	2023.0	2024.0	2025.0	2025.0	0.807551
Quarter	720.0	2.5125	1.0	2.0	3.0	3.0	4.0	1.106239

6.2 Trend Analysis

Trend analysis was performed to observe how healthcare capacity changed over time.

The line chart illustrates the variation in children under CBP custody and HHS care across different dates. The analysis revealed periods of increased healthcare demand followed by gradual stabilization.

This visualization helps decision-makers understand long-term operational behavior.

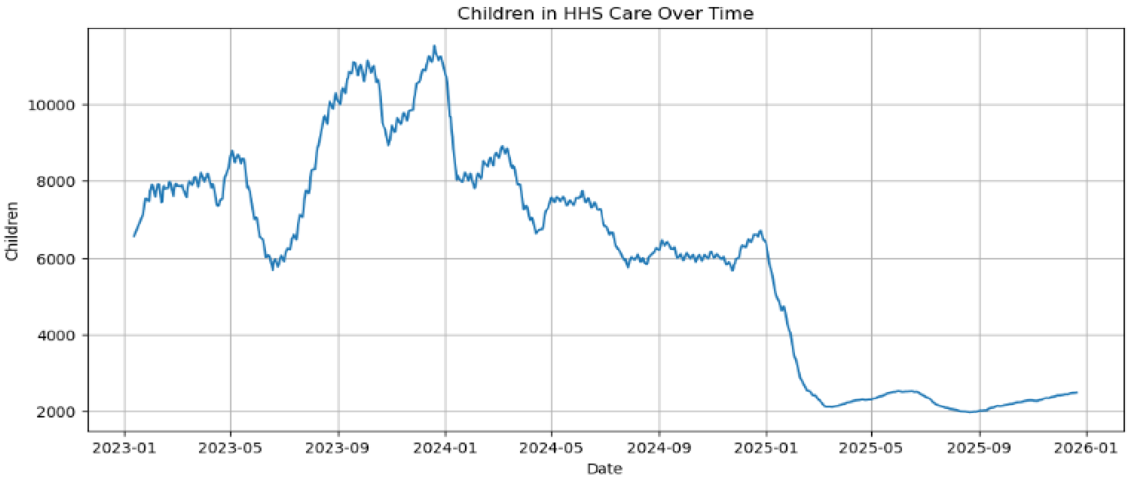

Figure 1: Trend Analysis

6.3 Distribution Analysis

A histogram was created to examine the distribution of healthcare capacity values.

The distribution indicates that most observations are concentrated within a specific range, while a few extreme values represent unusually high healthcare demand.

Distribution analysis assists in understanding data variability and identifying skewness.

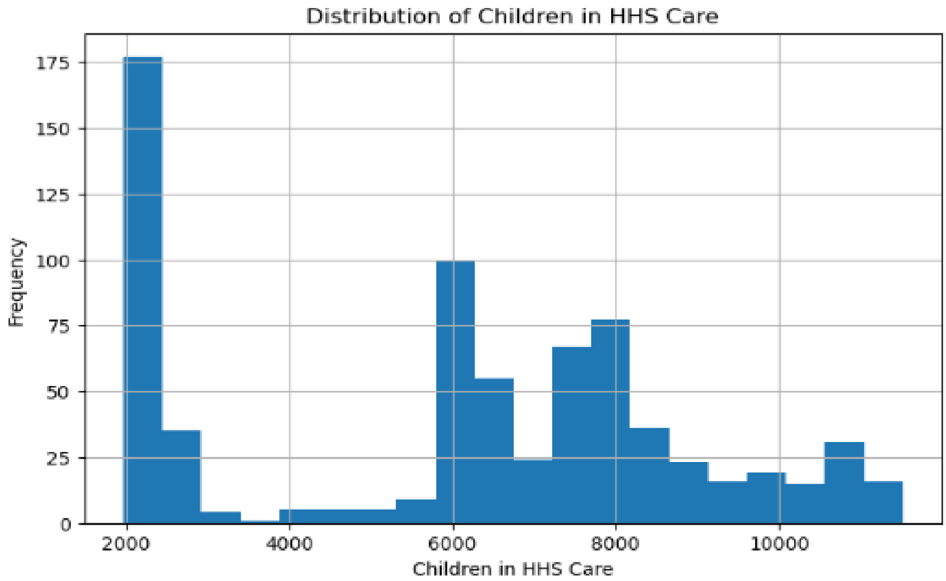

Figure 2: Histogram Distribution

6.4 Box Plot Analysis

A box plot was used to identify the spread of data and detect possible outliers.

The visualization highlights the median, quartiles, and extreme observations present in the healthcare dataset.

Outlier detection is useful for identifying unusual operational events that may require further investigation.

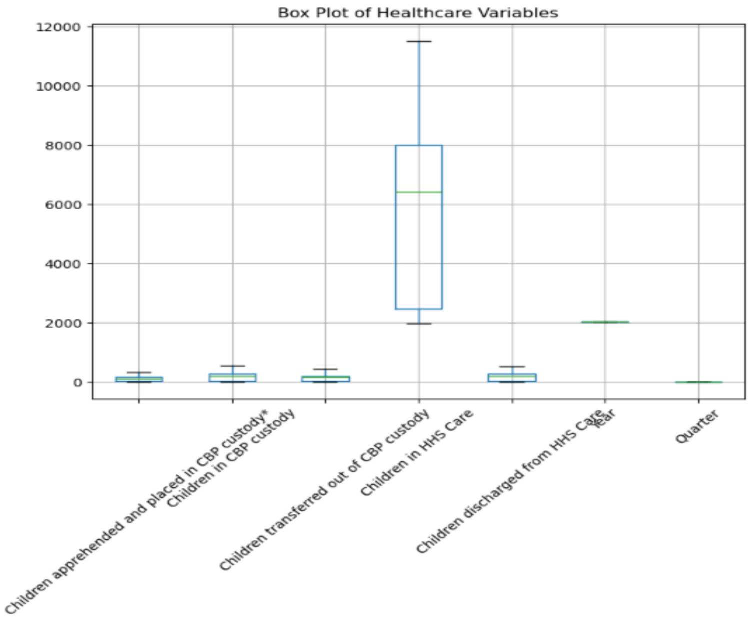

Figure 3: Box Plot

6.5 Correlation Analysis

Correlation analysis was performed to evaluate relationships among healthcare variables.

A correlation heatmap was generated to identify positive and negative relationships between key performance indicators such as:

- Children Apprehended
- CBP Custody
- HHS Care
- Transfers
- Discharges

Strong positive correlations indicate variables that increase together, while weak correlations suggest independent behavior.

	Children apprehended and placed in CBP custody*	Children in CBP custody	Children transferred out of CBP custody	Children in HHS Care	Children discharged from HHS Care	Year	Quarter
Children apprehended and placed in CBP custody*	1.000000	0.950604	0.887896	0.691312	0.627755	-0.587536	-0.054502
Children in CBP custody	0.950604	1.000000	0.925039	0.663662	0.602144	-0.542306	-0.081307
Children transferred out of CBP custody	0.887896	0.925039	1.000000	0.713899	0.657322	-0.589598	-0.078396
Children in HHS Care	0.691312	0.663662	0.713899	1.000000	0.920881	-0.872610	-0.008549
Children discharged from HHS Care	0.627755	0.602144	0.657322	0.920881	1.000000	-0.834188	-0.079626
Year	-0.587536	-0.542306	-0.589598	-0.872610	-0.834188	1.000000	-0.047660
Quarter	-0.054502	-0.081307	-0.078396	-0.008549	-0.079626	-0.047660	1.000000

6.6 Monthly Average Analysis

Monthly averages were calculated to evaluate seasonal variations in healthcare demand.

The analysis showed how healthcare capacity fluctuated across different months, helping identify peak operational periods.

Monthly trend analysis supports better resource planning and healthcare capacity management.

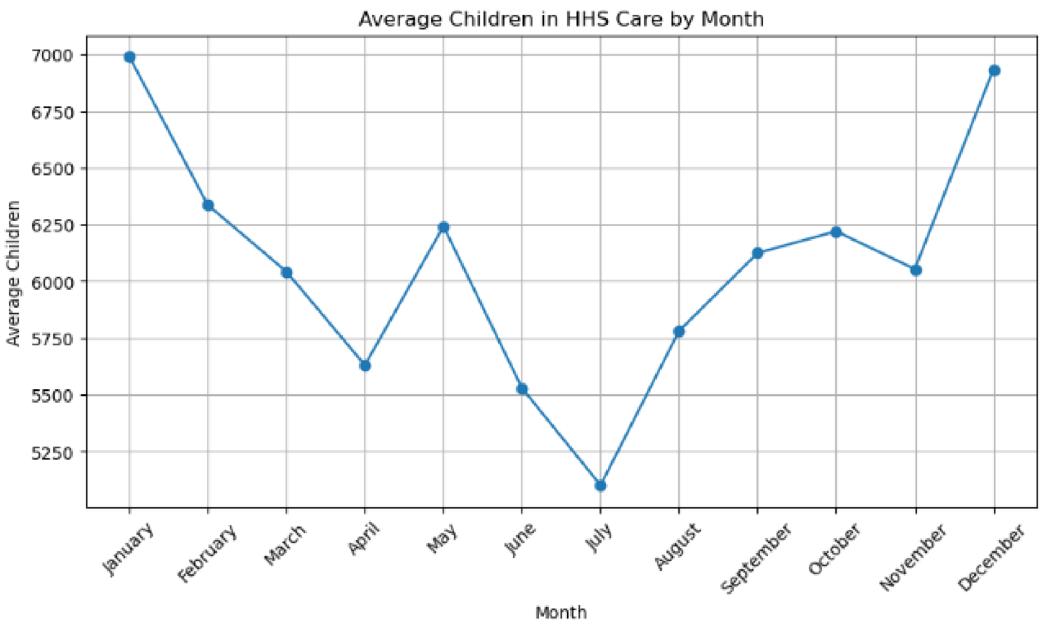

Figure 4: Monthly Analysis

6.7 Dataset Preview

A preview of the processed dataset was displayed within the Streamlit application.

This feature enables users to inspect records before performing further analysis.

It also improves transparency and provides quick access to the cleaned dataset.

Dataset Preview

	Date	Children apprehended and placed in CBP custody*	Children in CBP custody	Children transferred out of CBP custody	Children in HHS Care	Children discharged from HHS Care
719	2023-01-12 00:00:00	33	53	34	6566	436
718	2023-01-22 00:00:00	32	49	39	7122	227
717	2023-01-23 00:00:00	32	50	39	7280	181
716	2023-01-24 00:00:00	47	42	47	7433	175
715	2023-01-25 00:00:00	20	22	41	7538	180
714	2023-01-29 00:00:00	23	45	11	7472	303
713	2023-01-30 00:00:00	34	54	29	7743	196
712	2023-01-31 00:00:00	26	36	36	7803	158
711	2023-02-01 00:00:00	25	32	27	7903	231

Figure 5: Dataset Preview

6.8 Key Findings from EDA

The Exploratory Data Analysis produced several important observations:

- Healthcare demand varied significantly over the study period.
- Peak HHS care was observed during high operational periods.
- Strong relationships existed between CBP custody and HHS care.
- Monthly healthcare demand showed noticeable fluctuations.
- A few extreme observations (outliers) were identified.
- The cleaned dataset was suitable for dashboard development and forecasting analysis.

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7. Dashboard Development

This section presents the development of the interactive Healthcare Capacity and Care Load Analytics Dashboard using Python and Streamlit. The dashboard integrates exploratory data analysis (EDA), key performance indicators (KPIs), trend visualization, dataset preview, and forecasting techniques into a single user-friendly application. It enables healthcare administrators and decision-makers to monitor healthcare capacity efficiently through interactive visualizations.

7.1 Dashboard Overview

The dashboard was developed using the Streamlit framework to provide an interactive interface for healthcare data analysis. It combines multiple analytical components into a single application, allowing users to explore healthcare trends, monitor KPIs, and visualize forecasting results without requiring programming knowledge.

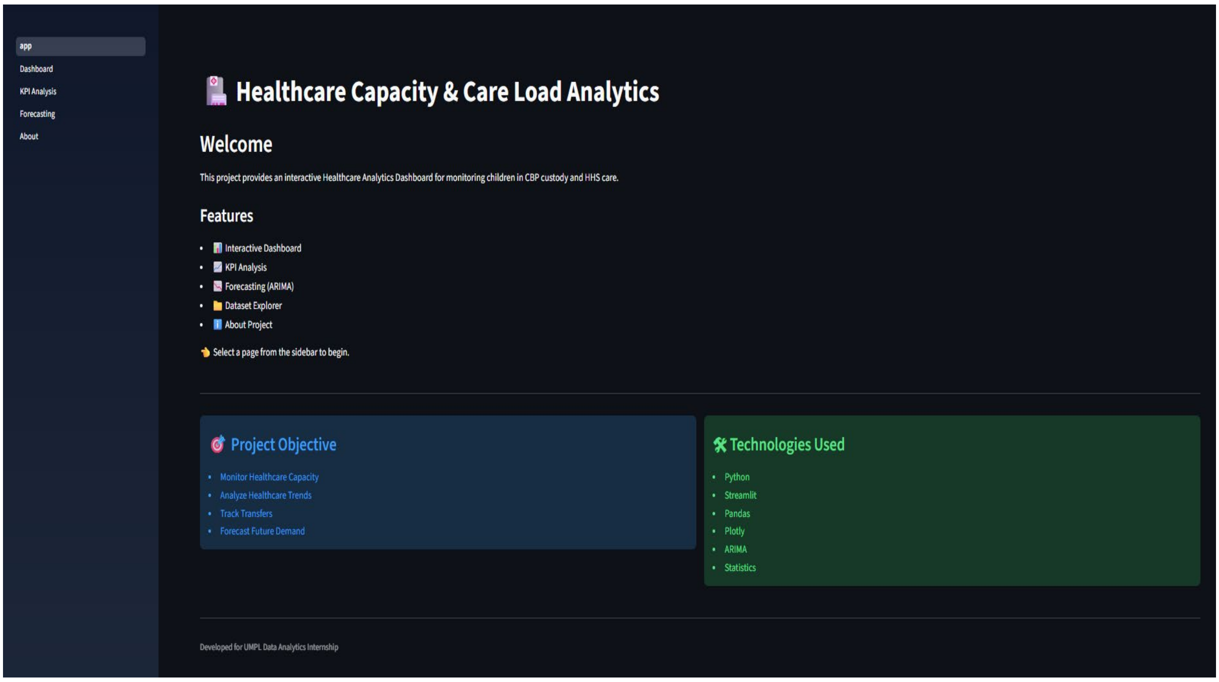
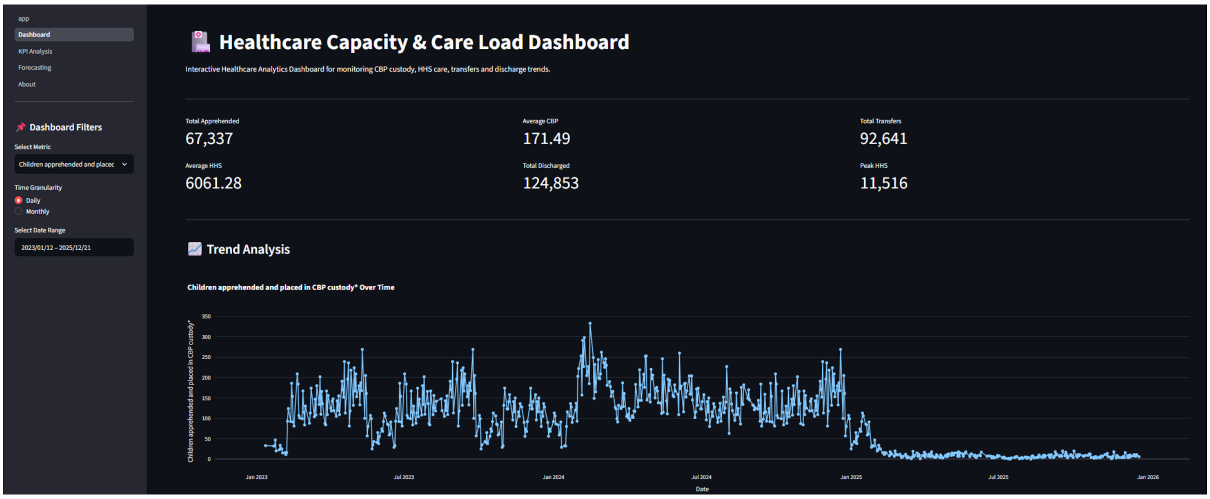
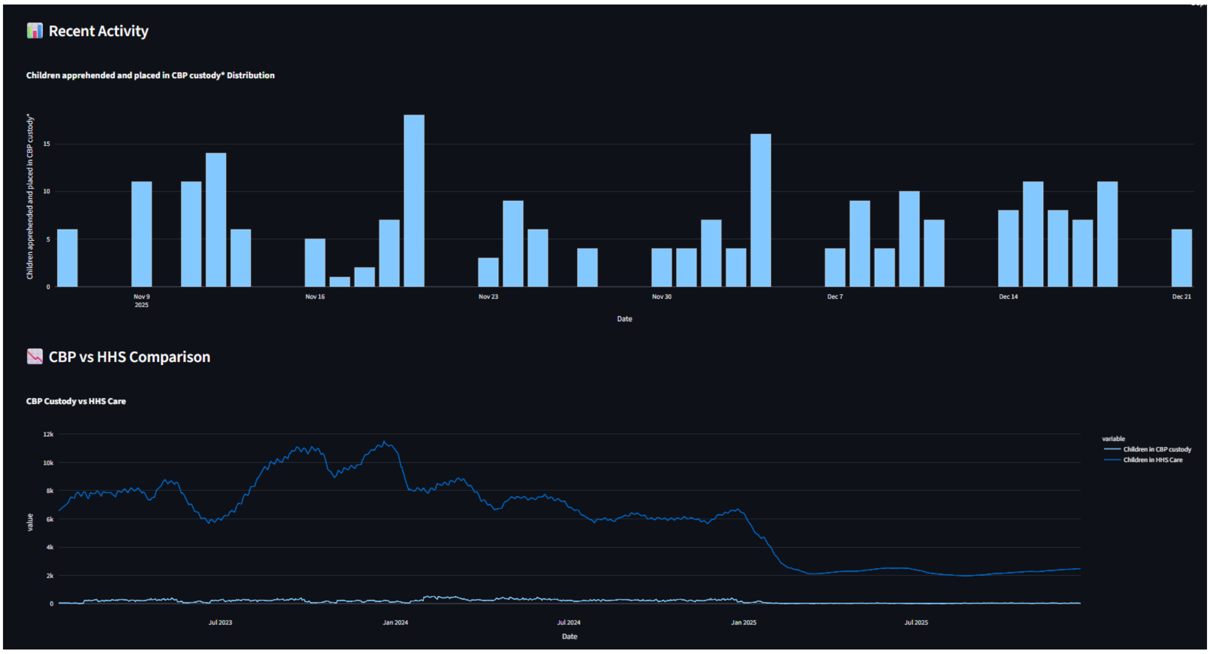


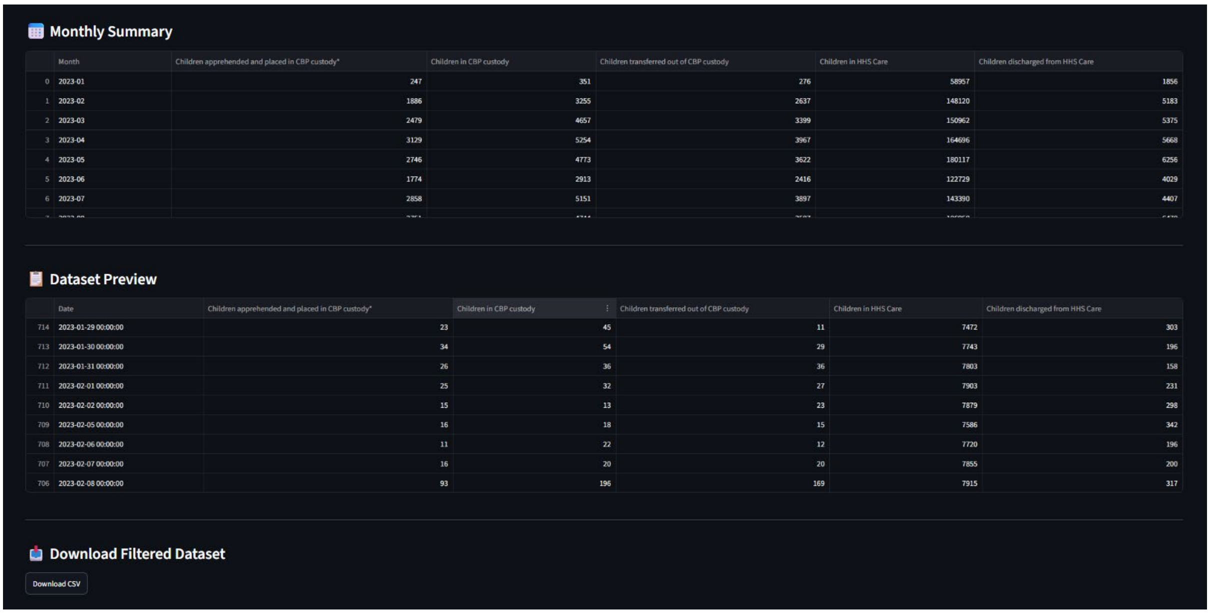
Figure 6: Dashboard Overview

7.2 Interactive Dashboard

The interactive dashboard presents important healthcare indicators using dynamic charts and visualizations. Users can easily analyze healthcare trends, compare operational metrics, and monitor variations across different healthcare variables.







Monthly Summary

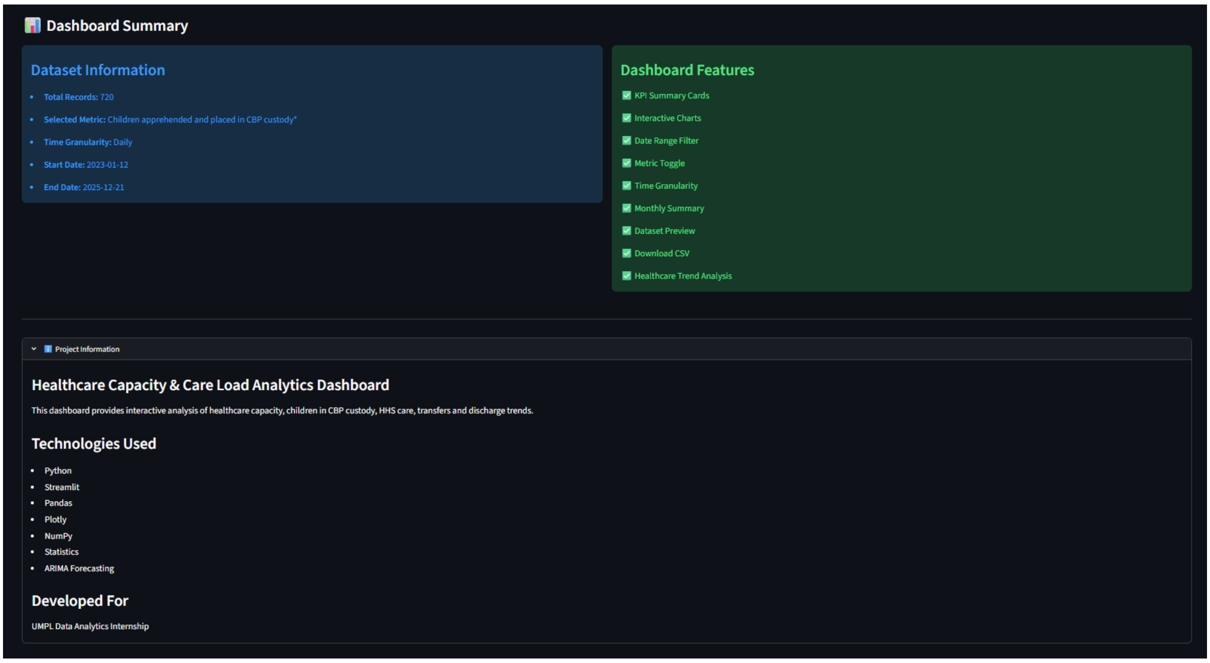
	Month	Children apprehended and placed in CBP custody*	Children in CBP custody	Children transferred out of CBP custody	Children in HHS Care	Children discharged from HHS Care
0	2023-01	247	351	276	58957	1856
1	2023-02	1886	3255	2637	148120	5183
2	2023-03	2479	4657	3399	150962	5375
3	2023-04	3129	5254	3967	164696	5668
4	2023-05	2746	4773	3622	180117	6256
5	2023-06	1774	2913	2416	122729	4029
6	2023-07	2858	5151	3897	143390	4407

Dataset Preview

	Date	Children apprehended and placed in CBP custody*	Children in CBP custody	Children transferred out of CBP custody	Children in HHS Care	Children discharged from HHS Care
714	2023-01-29 00:00:00	23	45	11	7472	303
713	2023-01-30 00:00:00	34	54	29	7743	196
712	2023-01-31 00:00:00	26	36	36	7803	158
711	2023-02-01 00:00:00	25	32	27	7903	231
710	2023-02-02 00:00:00	15	13	23	7879	298
709	2023-02-05 00:00:00	16	18	15	7586	342
708	2023-02-06 00:00:00	11	22	12	7720	196
707	2023-02-07 00:00:00	16	20	20	7855	200
706	2023-02-08 00:00:00	93	196	169	7915	317

Download Filtered Dataset

Download CSV



Dashboard Summary

Dataset Information

- **Total Records:** 720
- **Selected Metric:** Children apprehended and placed in CBP custody*
- **Time Granularity:** Daily
- **Start Date:** 2023-01-12
- **End Date:** 2025-12-21

Dashboard Features

- KPI Summary Cards
- Interactive Charts
- Date Range Filter
- Metric Toggle
- Time Granularity
- Monthly Summary
- Dataset Preview
- Download CSV
- Healthcare Trend Analysis

Project Information

Healthcare Capacity & Care Load Analytics Dashboard

This dashboard provides interactive analysis of healthcare capacity, children in CBP custody, HHS care, transfers and discharge trends.

Technologies Used

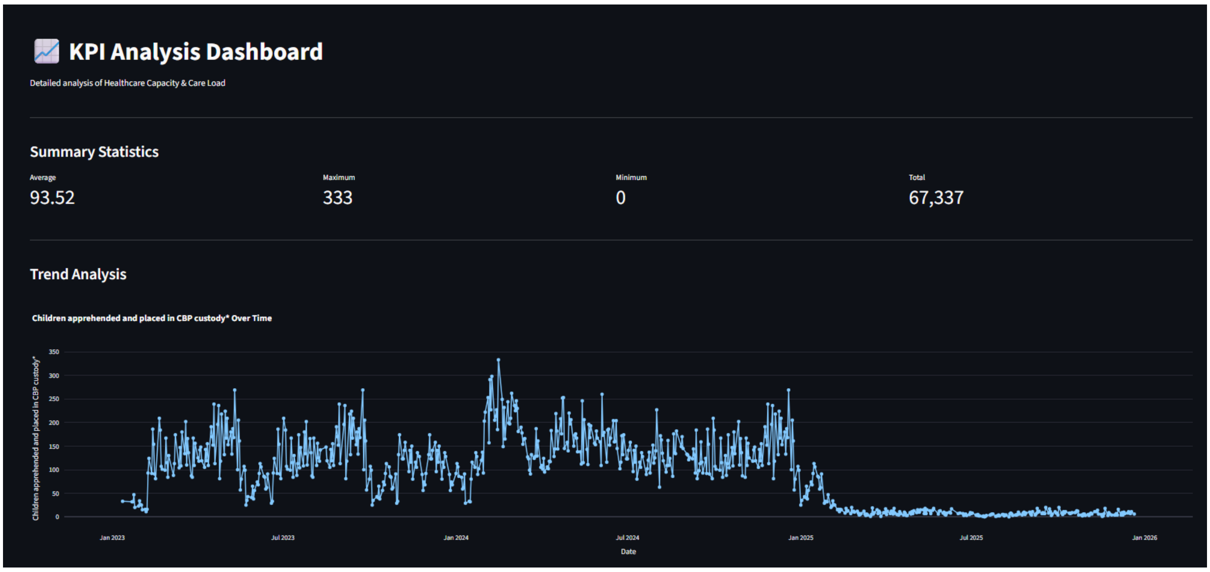
- Python
- Streamlit
- Pandas
- Plotly
- NumPy
- Statistics
- ARIMA Forecasting

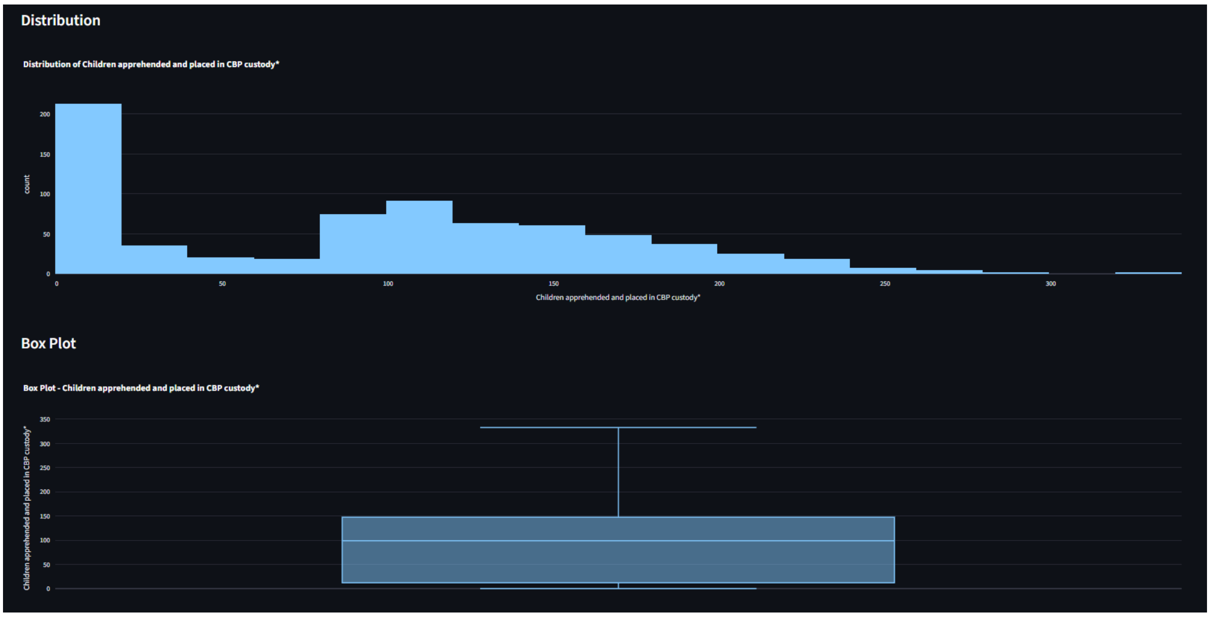
Developed For

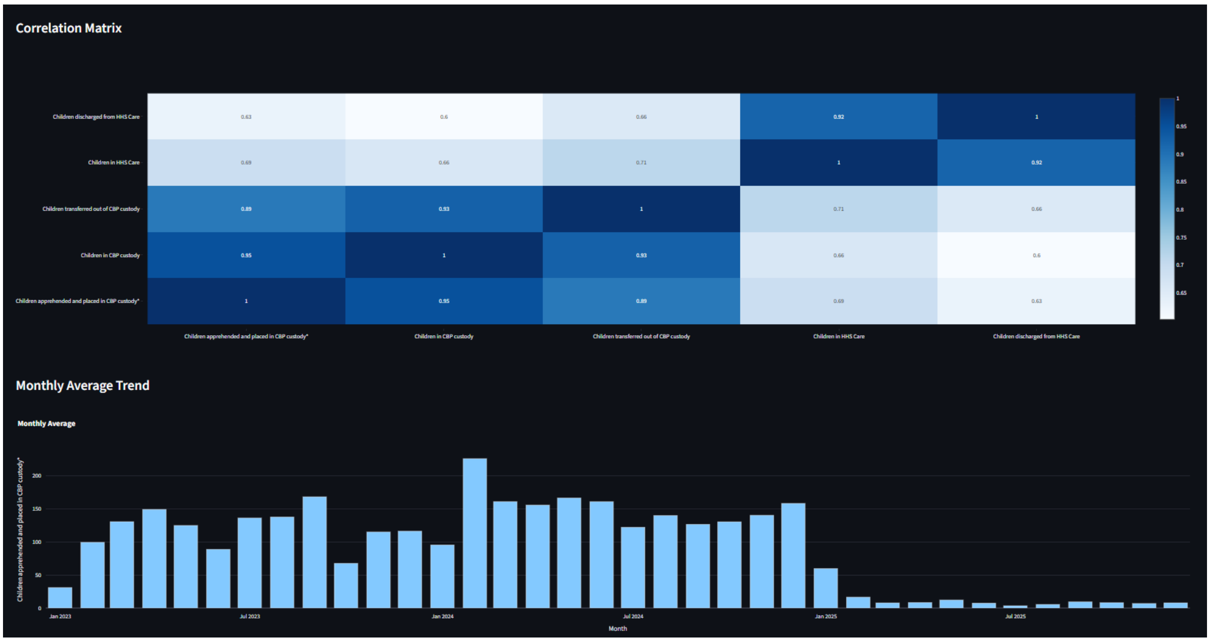
UMPL Data Analytics Internship

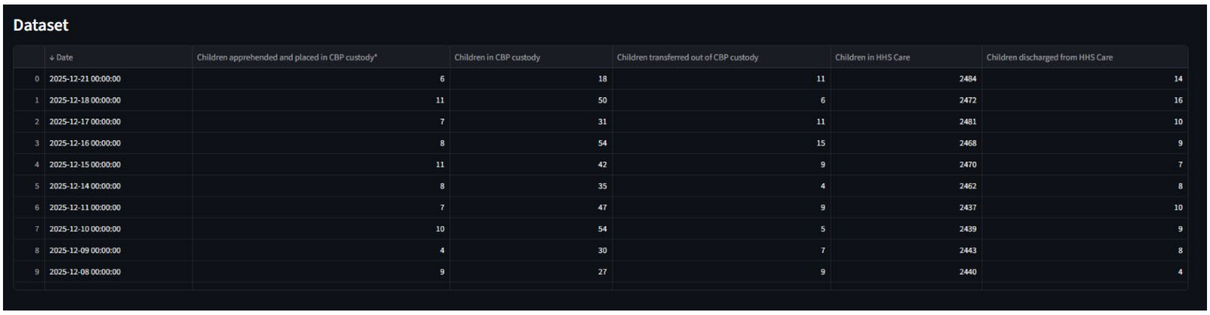
7.3 Key Performance Indicators (KPIs)

The dashboard displays key performance indicators such as Children in CBP Custody, Children in HHS Care, Transfers, and Discharges. These KPIs provide a quick overview of healthcare operational performance and assist administrators in monitoring critical healthcare metrics.









Dataset

	Date	Children apprehended and placed in CBP custody*	Children in CBP custody	Children transferred out of CBP custody	Children in HHS Care	Children discharged from HHS Care
0	2025-12-21 00:00:00	6	18	11	2484	14
1	2025-12-18 00:00:00	11	50	6	2472	16
2	2025-12-17 00:00:00	7	31	11	2481	10
3	2025-12-16 00:00:00	8	54	15	2468	9
4	2025-12-15 00:00:00	11	42	9	2470	7
5	2025-12-14 00:00:00	8	35	4	2462	8
6	2025-12-11 00:00:00	7	47	9	2437	10
7	2025-12-10 00:00:00	10	54	5	2439	9
8	2025-12-09 00:00:00	4	30	7	2443	8
9	2025-12-08 00:00:00	9	27	9	2440	4

Figure 7: KPI Dashboards

7.4 Forecasting Dashboard

An ARIMA-based forecasting model was integrated into the dashboard to predict future healthcare demand. The forecasting visualization helps decision-makers estimate future capacity requirements and supports proactive healthcare planning.

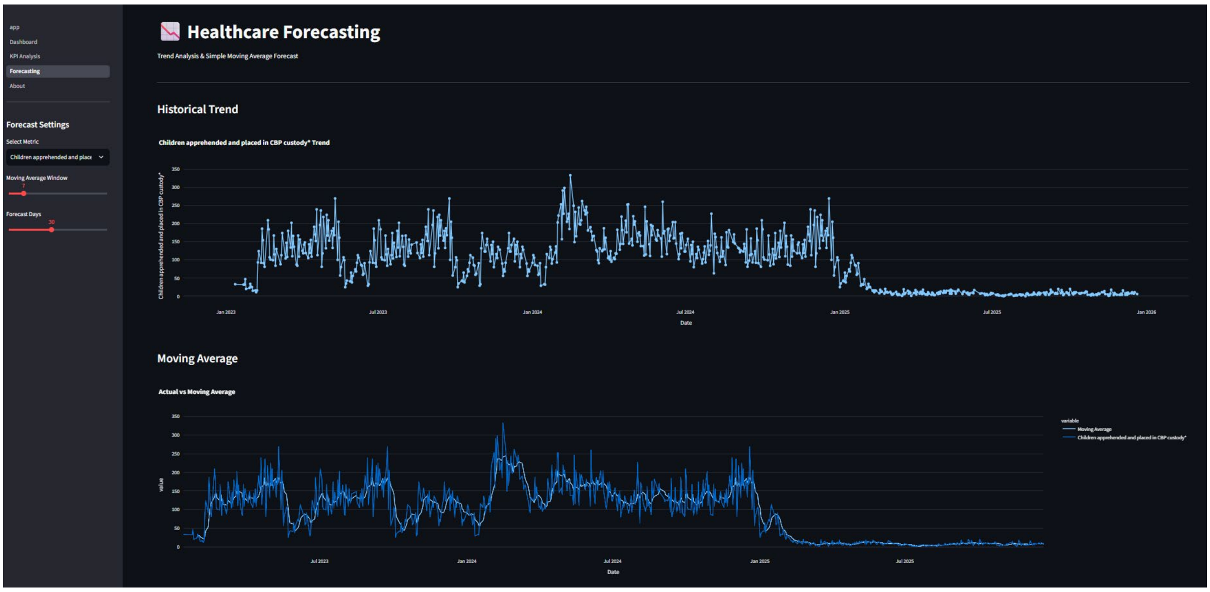
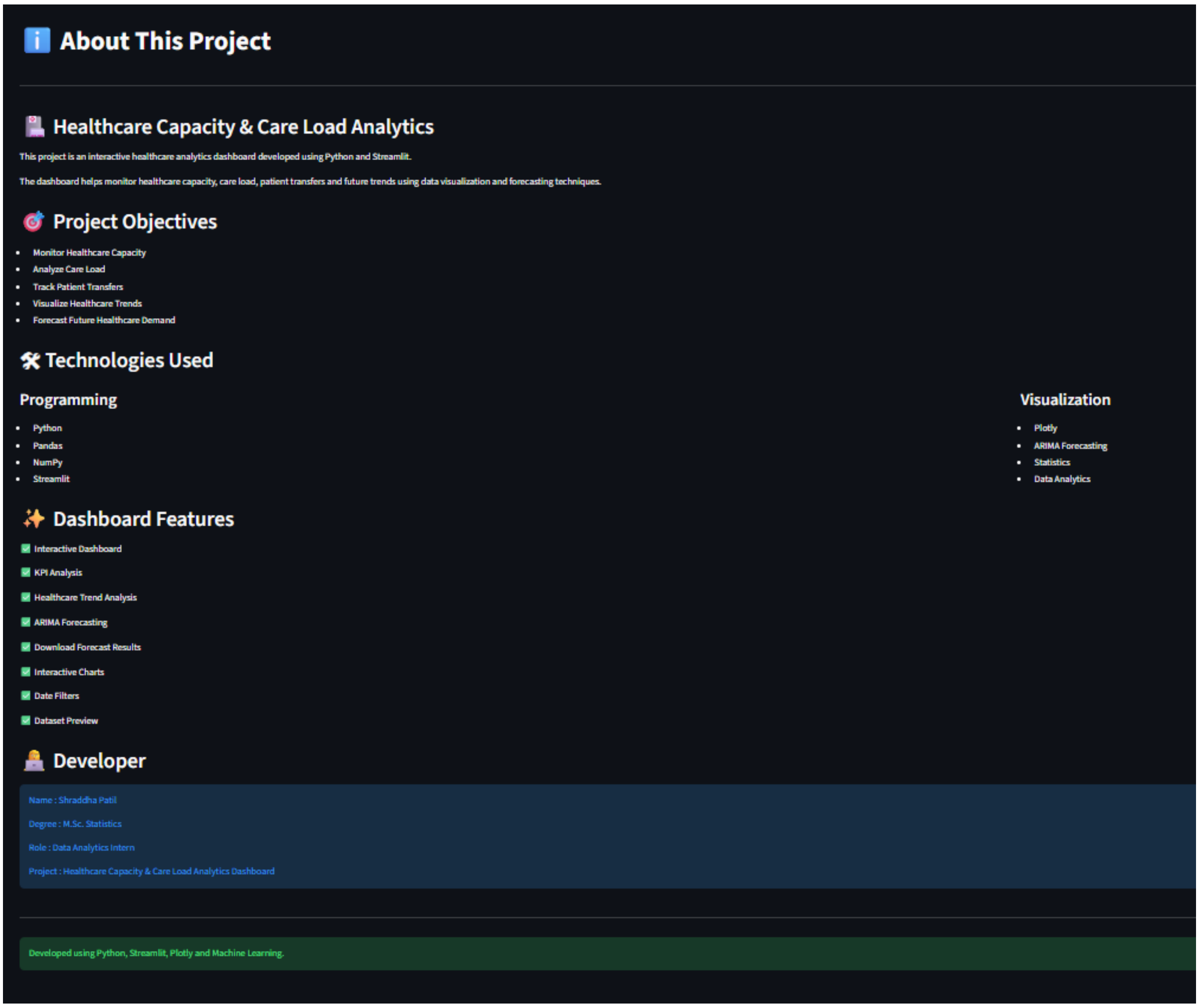


Figure 8: Forecast Trend

7.5 About Dashboard

The dashboard includes an About section that provides information about the project objectives, dataset, analytical techniques, and technologies used during development. This improves usability and helps users understand the purpose of the application.



7.6 Dashboard Features

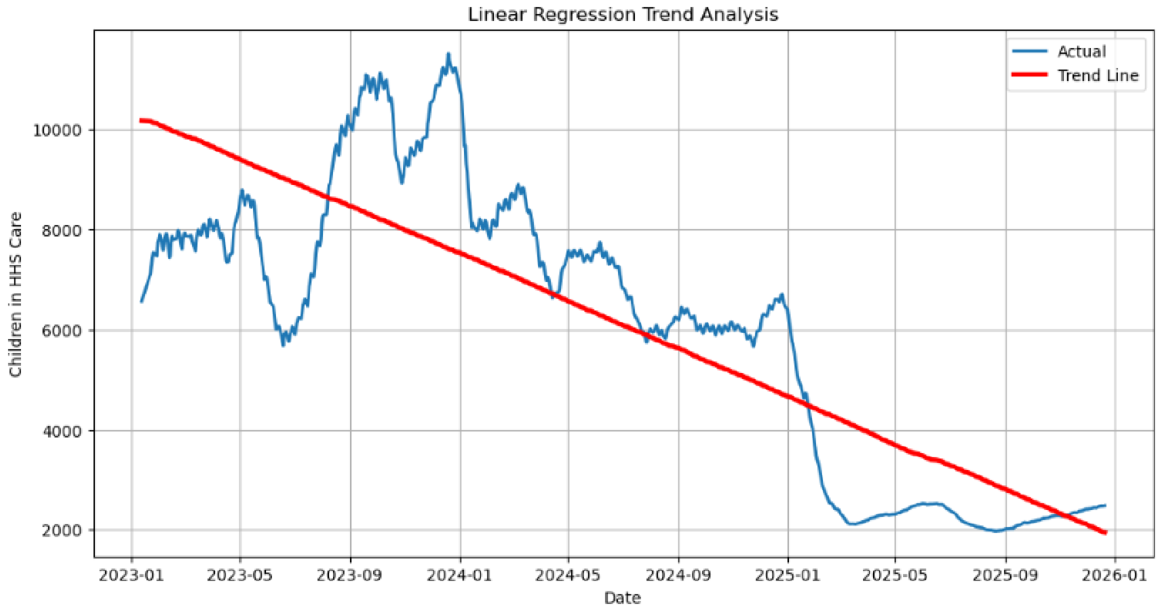
The developed dashboard provides several useful features for healthcare analytics:

- Interactive Streamlit web interface
- Real-time dataset preview
- Exploratory Data Analysis (EDA)
- KPI monitoring
- Trend analysis using visualizations
- Distribution and outlier analysis
- Correlation analysis
- ARIMA-based forecasting
- User-friendly navigation
- Fast and responsive dashboard performance

8. Forecasting Analysis

8.1 Linear Regression Trend Analysis

Forecasting is an important analytical technique used to estimate future healthcare demand based on historical observations. In this project, Linear Regression was applied to identify the overall trend in the number of children receiving HHS care. The model learns the relationship between time and healthcare demand, providing a simple yet effective forecasting approach for long-term trend analysis.



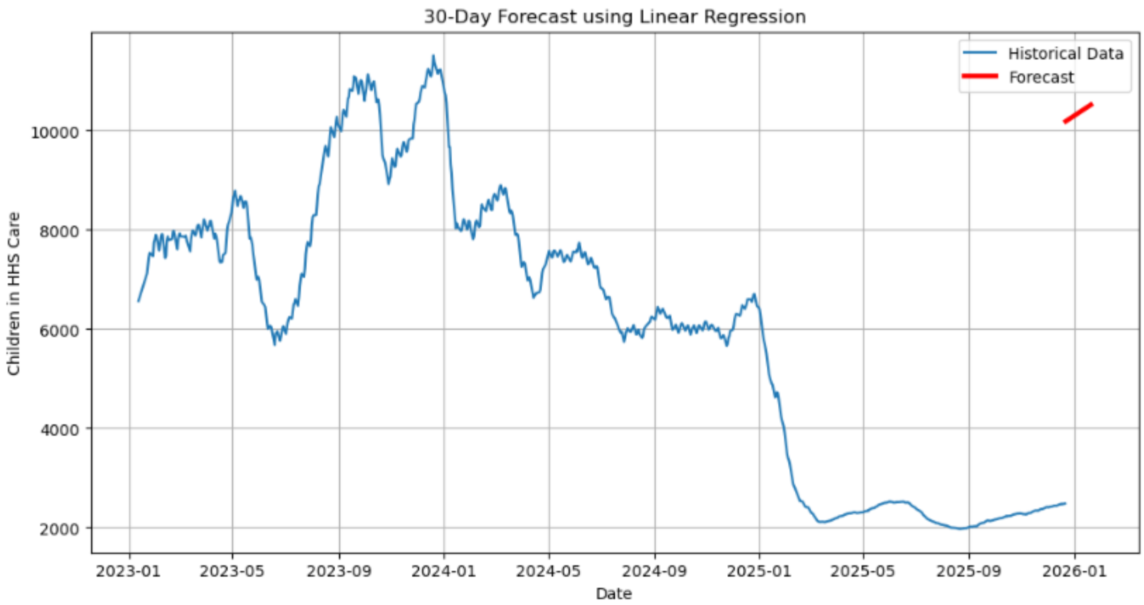
Linear Regression Trend Analysis

Observation

The Linear Regression trend line closely follows the historical healthcare data, highlighting the overall direction of healthcare demand. Although short-term fluctuations are present in the original data, the trend line clearly illustrates the long-term movement. This analysis indicates that healthcare demand remains relatively stable throughout the observed period, making it useful for monitoring healthcare capacity.

8.2 Future Healthcare Demand Forecast

After training the Linear Regression model, future healthcare demand was predicted for the next 30 days. The forecasting process estimates future values based on historical patterns observed in the dataset. These predictions can support healthcare administrators in planning resources, workforce allocation, and future healthcare services.



30-Day Forecast using Linear Regression

Observation

The forecasted values indicate a smooth continuation of the historical trend with no significant spikes or sudden declines. The predicted healthcare demand remains relatively consistent during the forecast period. Such predictions can help healthcare organizations prepare for future service requirements and improve operational planning.

Forecasted Healthcare Demand (First 10 Records)

```
[54]: forecast_result.head(10)
```

[54]:

	Date	Forecast
0	2025-12-22	10188.973099
1	2025-12-23	10200.423025
2	2025-12-24	10211.872950
3	2025-12-25	10223.322875
4	2025-12-26	10234.772801
5	2025-12-27	10246.222726
6	2025-12-28	10257.672651
7	2025-12-29	10269.122577
8	2025-12-30	10280.572502
9	2025-12-31	10292.022428

Figure 9: Forecast Table

Forecast Summary

The Linear Regression model successfully generated future healthcare demand predictions using historical healthcare records. Although it is a simple forecasting technique, it effectively demonstrates the application of predictive analytics in healthcare capacity planning. More advanced forecasting models such as ARIMA, Prophet, or LSTM can be explored in future work to improve prediction accuracy.

9. Results and Discussion

The developed Healthcare Capacity and Care Load Analytics system successfully analyzed historical healthcare records using Python and Streamlit. The Exploratory Data Analysis (EDA) identified important trends, seasonal variations, and relationships among healthcare variables.

The interactive dashboard provided real-time visualization of key healthcare indicators, including KPI cards, trend analysis, correlation heatmap, distribution analysis, and dataset exploration. These visualizations simplified data interpretation and supported better understanding of healthcare capacity.

Linear Regression forecasting was performed to estimate future healthcare demand. The generated trend line and 30-day forecast demonstrated the potential of predictive analytics for healthcare planning. The forecasting results indicated a relatively stable demand pattern, which can assist healthcare administrators in planning resources and improving decision-making.

Overall, the developed system successfully combines data analysis, visualization, and forecasting into a single interactive platform for healthcare capacity monitoring.

Key Findings

- Successfully performed data preprocessing and cleaning.
- Conducted comprehensive Exploratory Data Analysis (EDA).
- Developed an interactive Streamlit dashboard with filtering options.
- Visualized healthcare trends using charts and KPIs.
- Applied Linear Regression for trend analysis and future forecasting.
- Generated a 30-day healthcare demand forecast.
- Demonstrated the usefulness of predictive analytics for healthcare planning.

10. Conclusion

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This project successfully developed a **Healthcare Capacity and Care Load Analytics** system using Python, Streamlit, and data visualization techniques. The project involved data preprocessing, exploratory data analysis, dashboard development, and forecasting to analyze healthcare demand trends.

The interactive dashboard enables users to monitor healthcare indicators, explore historical patterns, and visualize important performance metrics through an intuitive interface. Linear Regression forecasting further enhanced the project by providing estimates of future healthcare demand based on historical observations.

The proposed system supports data-driven decision-making, efficient healthcare resource planning, and improved operational management. In future work, advanced forecasting models such as ARIMA, Facebook Prophet, or LSTM can be implemented to improve prediction accuracy and support real-time healthcare analytics.

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